Cranberry Models: Common Control

Soren Basnet, Qingquan Liu, Wuyang Wang

1) Introduction

Crop production and harvest affect everyone, because everyone, regardless of their circumstances, needs food to survive. As global temperatures rise, food production and harvest quantity may fluctuate.

This study aims to answer questions about the effects of heat treatment on cranberries and the resulting changes in cranberry production.

We aim of answer the following questions :

1. How does heat stress affect yield as measured by the total weight of fruit?
2. How does heat stress affect yield as measured by the total weight of healthy fruit?
3. How does heat stress affect yield as measured by the percentage of healthy berries by count?
4. How does heat stress affect yield as measured by the percentage of healthy fruit by weight?
5. How do the cultivars, years, and intensity of the heat shocks (in the case of the acute heat stress experiment) affect the various measures of yield?

2) Data and Experimental Design

Experimental design. The data contain 96 plots of two cranberry cultivars, Mullica Queen and Stevens, each grown in its own field, in 2024 and 2025. The plots form 48 pairs of neighbouring plots: in each pair, one plot was heated and the other was left unheated as a Control. Heated plots received 1, 2, 3 or 4 heat pulses (intensity groups D, C, B and A in the data), with 3 pairs for each number of pulses in every combination of cultivar and year. All Control plots received the same treatment, whatever happened to their paired heated plot, so we treat the 12 Control plots of each cultivar and year as one Control group. The heat treatment therefore has five levels: Control and 1, 2, 3 or 4 pulses.

Yield measures. For each plot, the data record the number and weight of healthy and rotten berries. From these we calculated the four measures of yield studied in this report: total fruit weight (healthy plus rotten weight), healthy fruit weight, the percentage of healthy berries by count, and the percentage of healthy fruit by weight.

Data checks. The data have no missing values. Every pair contains exactly one Control and one heated plot, and all weights and berry counts are positive. No observations were removed, so all analyses use the full 96 plots.

Paired design. The two plots of a pair are neighbours and share soil and growing conditions, so their yields are not independent. All models in this report therefore include a random effect for each pair, which allows each pair to have its own baseline yield.

First look at the data. Control yields were similar in three of the four combinations of cultivar and year, but Stevens produced only about a third as much fruit in 2025 (Table 1). To put all plots on a common scale, Figure 1 expresses each plot as a percentage of the mean Control value of its own cultivar and year, with both cultivars shown together. Most heated groups had a lower average yield than Control. For the two weights, the clearest reductions occurred with three or four pulses in 2025, whereas in 2024 there was no clear pattern with the number of pulses. For the two percentages, the share of healthy fruit was lower

in every heated group, by a broadly similar amount whatever the number of pulses. Plots varied widely within each group, and many heated plots overlapped with Control plots, so the models in Sections 3 to 6 are needed to judge which differences are reliable.

Table 1: Mean of each yield measure in the Control plots, by cultivar and year.

Cultivar	Year	Total (g)	Healthy (g)	Healthy, % by count	Healthy, % by weight
Mullica Queen	2024	1091	1060	95.8	97.1
Mullica Queen	2025	1096	1057	94.5	96.4
Stevens	2024	961	945	95.6	98.4
Stevens	2025	379	374	95.2	98.9

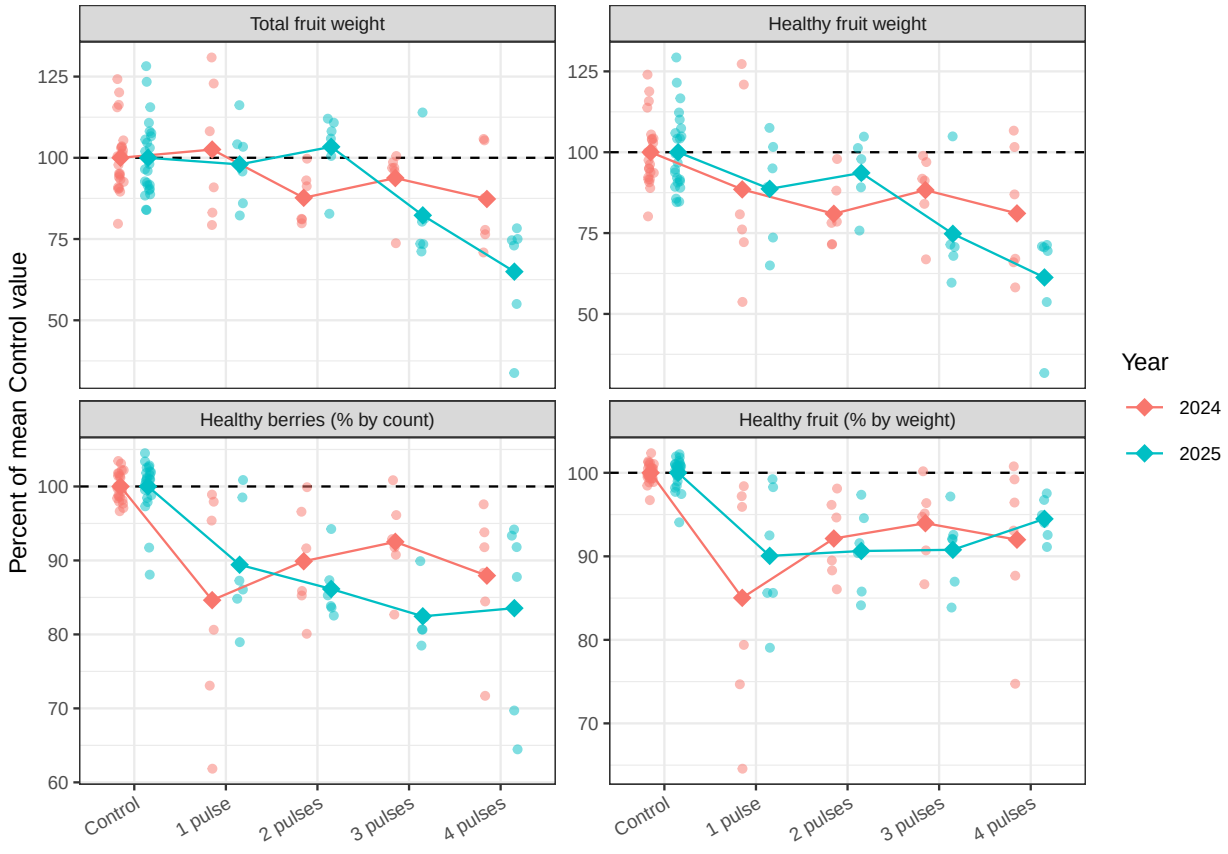


Figure 1: Each yield measure as a percentage of the mean of the Control plots of the same cultivar and year, with both cultivars combined. Small points are individual plots; diamonds joined by lines are group means, coloured by year. The dashed line at 100 marks the Control level. Vertical scales differ between panels.

3) How does heat stress affect yield as measured by the total weight of fruit?

Appendix A compares the candidate models for total weight. The fitted model has heat group by year, cultivar, and the cultivar-by-year shift in its fixed part, a random intercept for each pair, and separate residual variances for Control and heated plots.

3.1 Final model

The final model for plot i is

$$\log(Y_i) = \mu + \alpha_{g(i)} + \beta_{t(i)} + (\alpha\beta)_{g(i)\,t(i)} + \gamma_{c(i)} + (\beta\gamma)_{t(i)\,c(i)} + b_{k(i)} + \varepsilon_i,$$

where Y_i is the total fruit weight in grams and log is the natural logarithm. μ is the overall intercept, α_g the effect of heat group g, β_t the effect of year t, γ_c the effect of cultivar c, $(\alpha\beta)_{gt}$ the heat-by-year interaction, and $(\beta\gamma)_{tc}$ the cultivar-by-year shift. b_k is the random intercept of pair k, with $b_k \sim N(0, \sigma_b^2)$, and ε_i is the residual error, with variance $\sigma^2_{\text{Control}}$ in Control plots and σ^2_{Heated} in heated plots. The estimated standard deviations are $\hat{\sigma}_b = 0.067$, $\hat{\sigma}_{\text{Control}} = 0.09$ and $\hat{\sigma}_{\text{Heated}} = 0.164$. The model was fitted by restricted maximum likelihood in R with nlme.

3.2 Coefficients of the final model

Table 2: Fixed-effect estimates of the final heteroscedastic H2 model.

Term	Estimate	SE	p_value	Ratio	Lower_95	Upper_95
(Intercept)	6.987	0.031	< 1e-04	1082.535	1016.681	1152.654
heat_groupD	-0.006	0.072	0.93396	0.994	0.859	1.150
heat_groupC	-0.140	0.072	0.05976	0.870	0.752	1.006
heat_groupB	-0.044	0.072	0.54507	0.957	0.827	1.107
heat_groupA	-0.135	0.072	0.06883	0.874	0.755	1.011
year2025	-0.012	0.044	0.79197	0.988	0.904	1.080
cultivarST	-0.121	0.042	0.00636	0.886	0.814	0.965
heat_groupD:year2025	0.001	0.102	0.99550	1.001	0.814	1.230
heat_groupC:year2025	0.160	0.102	0.12503	1.173	0.955	1.442
heat_groupB:year2025	-0.167	0.102	0.10883	0.846	0.688	1.040
heat_groupA:year2025	-0.321	0.102	0.00311	0.725	0.590	0.891
year2025:cultivarST	-0.906	0.060	< 1e-04	0.404	0.358	0.456

Control, 2024 and Mullica Queen are the reference levels. A `heat_group` coefficient is the log ratio of total weight in that heat group to Control in 2024 for Mullica Queen. A `:year2025` term is the additional change in 2025, `cultivarST` is the Stevens minus Mullica Queen difference in 2024, and `year2025:cultivarST` is the extra 2025 change for Stevens.

3.3 Predicted yields

The figure below shows the fitted mean total fruit weight for each heat group, cultivar and year, with 95% confidence intervals. In 2024 the four heat groups sit close to Control. In 2025 the estimate falls from Control to four pulses. Mullica Queen has the higher baseline, so the same proportional fall is the larger loss in grams.

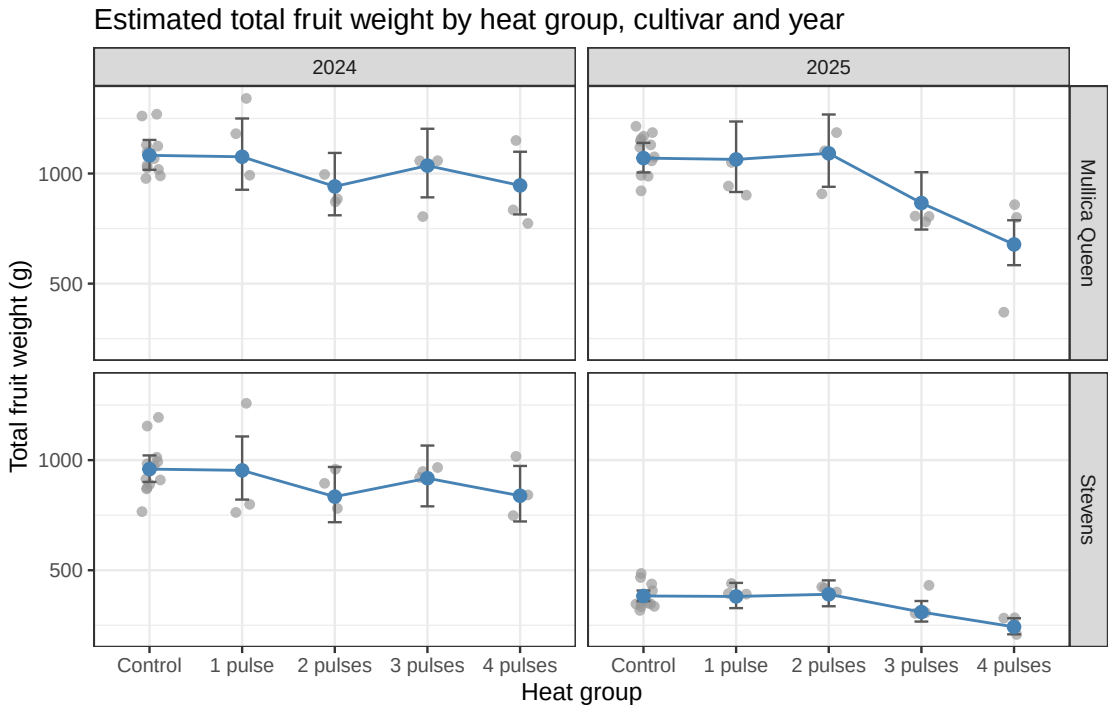


Figure 2: Estimated total fruit weight by heat group, cultivar (rows) and year (columns). Blue points are the means estimated by the final model, joined by lines, with 95% confidence intervals shown as bars; grey points are the individual plots. Control and heated plots are allowed different residual variances, so the intervals of heated plots are wider.

3.4 Answer

Heat stress lowers total fruit weight, and the size of the drop depends on the year and on the number of pulses. Averaged over the four heat groups, heated plots produced about 11.5% less fruit than Control. The reduction was about 7.8% in 2024 and about 15.1% in 2025. Within 2025, four pulses gave about 36.6% less fruit than Control and three pulses about 19.1% less. Across the four heat groups the estimate falls by about 9% for each additional pulse, but the slope is weak in 2024 at about -3% per pulse ($p = 0.36$) and strong in 2025 at about -15% per pulse ($p < 0.001$). There is no clear heat-by-cultivar interaction ($p = 0.28$): Mullica Queen and Stevens differ mainly in their baseline, and the Stevens baseline is much lower in 2025. Control and heated plots were also allowed different residual variances, which the likelihood-ratio test supports, so the estimates above come from that model.

4) How does heat stress affect yield as measured by the total weight of healthy fruit?

4.1 First look

Comparing each heated plot with its own Control. Figure 3 shows the ratio of healthy fruit weight between each heated plot and its neighbouring Control plot. In 37 of the 48 pairs (77%), the heated plot produced less healthy fruit than its Control, so heat reduced healthy weight in most pairs. The pattern across heat intensities differed between years. In 2024, the ratios were similar for all four intensities: in most groups, heated plots produced about 15 to 25% less healthy fruit than their Controls, and there was no tendency for more pulses to cause a larger reduction. In 2025, the ratios decreased as the number of heat

pulses increased, and heated plots that received three or four pulses produced clearly less healthy fruit than those that received one or two.

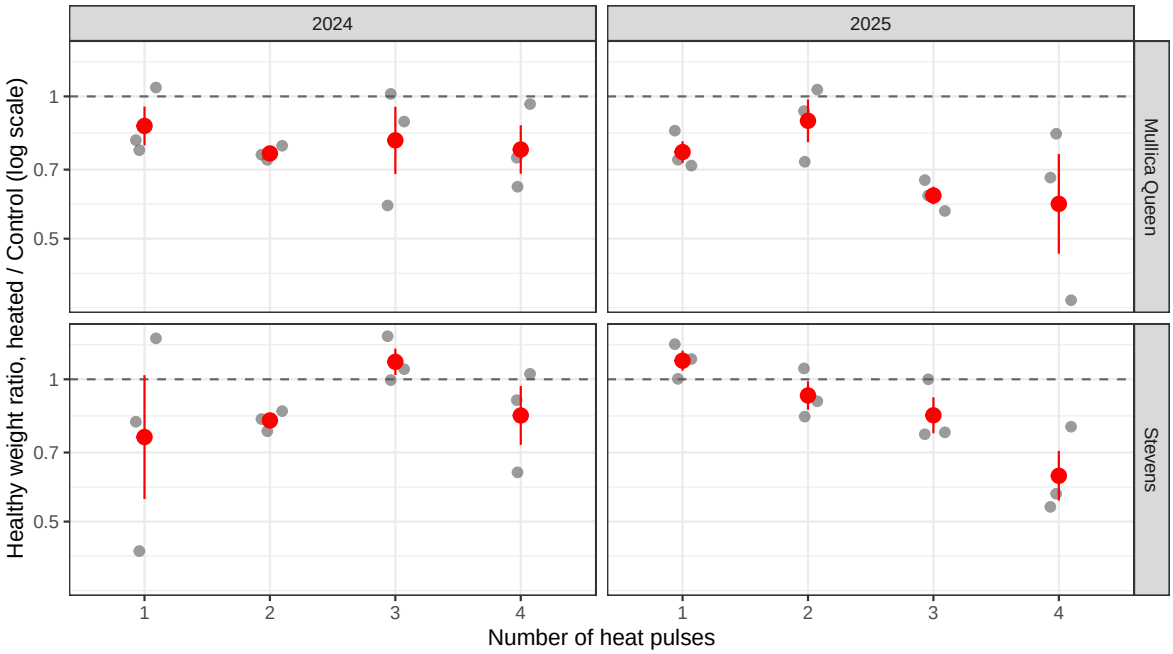


Figure 3: Ratio of healthy fruit weight in each heated (OTC) plot to that in its paired Control plot, by number of heat pulses, cultivar (rows) and year (columns). Grey points are the 48 individual pairs; red points and bars show the mean and plus or minus one standard error of the log ratios. The dashed line marks a ratio of 1 (no difference); points below it are pairs in which the heated plot produced less healthy fruit than its own Control. The vertical axis is on a logarithmic scale.

4.2 Methods

We modelled the natural logarithm of healthy fruit weight with a linear mixed model. The fixed effects were heat group (Control or 1, 2, 3 or 4 pulses), year, cultivar, the interaction between heat group and year, and the interaction between year and cultivar. The model also included a random effect for each pair of plots and, because heated plots varied more than Control plots, a separate residual variance for heated and Control plots. It was fitted by restricted maximum likelihood with the nlme package in R. On the log scale, each heat effect is a proportional change, so we report it as the ratio of healthy weight in heated plots to that in Control plots, or as the corresponding percentage reduction. Within each year, we compared the average of the four heat groups with Control, compared each heat group with Control, adjusting for multiple comparisons with Dunnett's method, and summarised the effect of intensity as the average change per additional pulse among heated plots. We chose this model by comparing it with simpler and more complex alternatives: allowing the heat effect to differ between years improved the fit, whereas allowing it to differ between cultivars did not. Details of the model, its selection and its checks are given in Appendix B.

4.3 Results

Overall effect of heat. Averaged over the four heat groups, both cultivars and both years, heated plots produced 20% less healthy fruit than Control plots (95% CI 14% to 25%, $p < 0.001$). The average reduction was similar in the two years, 17% in 2024 and 22% in 2025 (Table 3).

Table 3: Estimated ratio of healthy fruit weight in heated plots to that in Control plots, averaged over the two cultivars; a ratio of 0.80 means 20% less healthy fruit. Results for individual heat groups are adjusted for multiple comparisons (Dunnett's method). The last row is the change per additional pulse from 1 to 4 pulses.

Comparison with Control	2024 ratio (95% CI)	p	2025 ratio (95% CI)	p
Average of the four heat groups	0.83 (0.76–0.92)	< 0.001	0.78 (0.71–0.85)	< 0.001
1 pulse	0.84 (0.66–1.06)	0.176	0.89 (0.71–1.12)	0.505
2 pulses	0.80 (0.64–1.01)	0.065	0.93 (0.74–1.17)	0.785
3 pulses	0.90 (0.71–1.13)	0.565	0.73 (0.58–0.93)	0.006
4 pulses	0.80 (0.63–1.00)	0.056	0.60 (0.47–0.75)	< 0.001
Per additional pulse	1.00 (0.92–1.08)	0.930	0.87 (0.80–0.94)	< 0.001

Individual heat groups and dose response. In 2025, the reduction grew with the number of pulses: plots that received three and four pulses produced 27% and 40% less healthy fruit than Control, and each additional pulse reduced healthy weight by about 13% (95% CI 6% to 20%). The estimated reductions for one and two pulses were smaller (11% and 7%), and their confidence intervals range from a reduction of about 29% to an increase of about 17%. In 2024, every heat group was estimated to produce 10% to 20% less healthy fruit than Control, with no trend with the number of pulses. None of these differences was statistically discernible on its own, but their confidence intervals range from reductions of up to about 37% to increases of up to about 13%, so reductions of 20% or more remain plausible at every intensity. The two years therefore differ mainly in how the effect depends on the number of pulses, rather than in its average size.

Cultivars. There was no evidence that the proportional effect of heat differed between Mullica Queen and Stevens, but because their baselines differ, the same proportional reduction means very different losses in grams (Figure 4). In 2025, for example, four pulses reduced the estimated healthy weight of Mullica Queen from 1037 g to 621 g, but that of Stevens only from 377 g to 225 g.

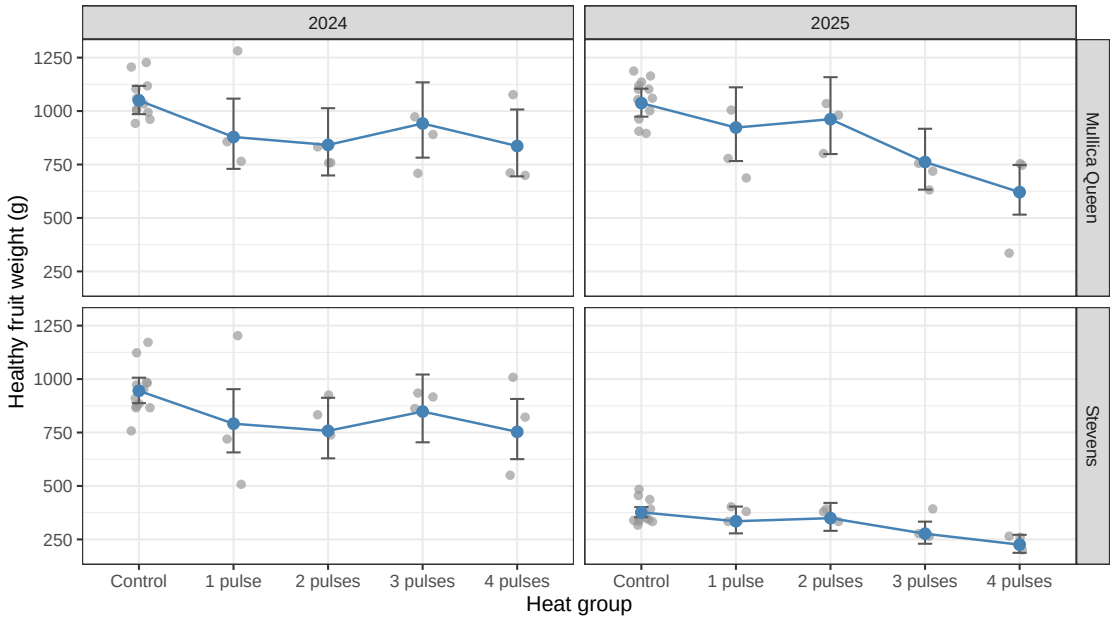


Figure 4: Estimated healthy fruit weight by heat group, cultivar (rows) and year (columns). Blue points are the means estimated by the final model, joined by lines, with 95% confidence intervals shown as bars; grey points are the individual plots. The model assumes that heat reduces healthy weight by the same proportion in both cultivars, so the estimated curves of the two cultivars have the same shape on a proportional scale.

Sensitivity to extreme plots. Removing the two heated plots with unusually low healthy weight made the estimated effects slightly smaller but did not change these conclusions. The only difference was that two heat groups in 2024 became individually statistically discernible, which confirms that the evidence for individual heat groups in 2024 is borderline (Appendix B.5).

4.4 Answer

Heat stress reduced the total weight of healthy fruit by about 20% on average (95% CI 14% to 25%), with a similar average reduction in 2024 and 2025. However, the effect of the number of pulses differed between years: in 2025 the reduction grew by about 13% with each additional pulse and reached about 40% with four pulses, whereas in 2024 the number of pulses made little difference. The proportional effect was similar for both cultivars, so losses in grams were larger for Mullica Queen, which has the higher baseline. Because this reduction is larger than that for total fruit weight (Section 3), heat appears to lower healthy weight partly by increasing the share of rotten fruit.

5) How does heat stress affect yield as measured by the percentage of healthy berries by count?

Heat changes both how many berries a plot carries and what share of them is healthy. This section looks at counts. The response is `healthy_n`, the number of healthy berries on a plot, which is a count, so the model is a negative-binomial GLMM with a log link, a random intercept for each pair, and all two-way interactions between heat group, cultivar and year.

5.1 Final model

The model is

```
healthy_n ~ (heat_group + cultivar + year)^2 + (1 | pair_id)
```

Because the link is logarithmic, a coefficient is a log ratio of expected counts, and `exp(beta)` is the multiplicative change in the expected number of healthy berries against the reference group, which is Control in 2024 for Mullica Queen. A ratio of 0.85 therefore means 15% fewer healthy berries.

5.2 Results

Averaged over the four heat groups, both years and both cultivars, heated plots produced 16% fewer healthy berries than Control, with a 95% confidence interval from 10% to 21% fewer.

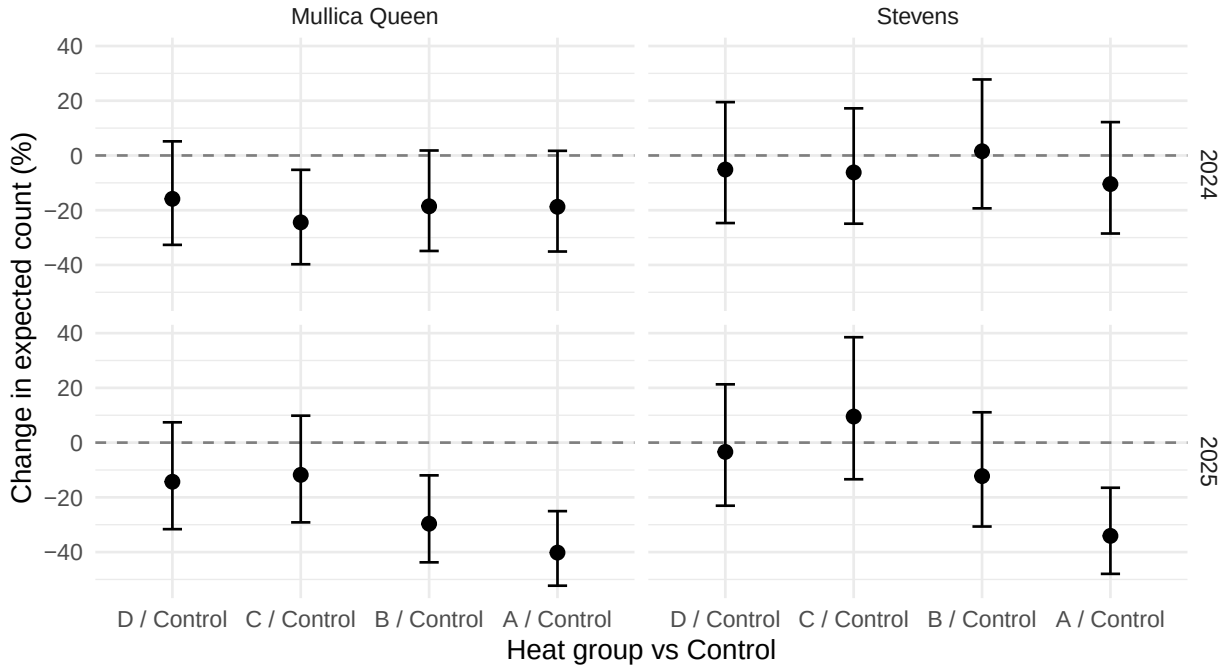


Figure 5: Change in the expected number of healthy berries relative to Control, by heat group, cultivar and year. Negative values are reductions. Bars are 95% confidence intervals from the final negative-binomial model, adjusted for four comparisons with the same Control.

Table 4: Effect of each heat group on the expected number of healthy berries, relative to Control, separately for each cultivar and year. Ratios are adjusted for four comparisons with the same Control using Dunnett's method. A ratio of 0.85 means 15% fewer healthy berries.

Cultivar	Year	Comparison	Ratio (95% CI)	Change in expected count	p-value
Mullica Queen	2024	1 pulse vs Control	0.84 (0.67 to 1.05)	-16%	0.182
Mullica Queen	2024	2 pulses vs Control	0.76 (0.60 to 0.95)	-24%	0.009
Mullica Queen	2024	3 pulses vs Control	0.81 (0.65 to 1.02)	-19%	0.082
Mullica Queen	2024	4 pulses vs Control	0.81 (0.65 to 1.02)	-19%	0.080
Mullica Queen	2025	1 pulse vs Control	0.86 (0.68 to 1.07)	-14%	0.277
Mullica Queen	2025	2 pulses vs Control	0.88 (0.71 to 1.10)	-12%	0.427
Mullica Queen	2025	3 pulses vs Control	0.70 (0.56 to 0.88)	-30%	< 0.001
Mullica Queen	2025	4 pulses vs Control	0.60 (0.48 to 0.75)	-40%	< 0.001
Stevens	2024	1 pulse vs Control	0.95 (0.75 to 1.19)	-5%	0.911
Stevens	2024	2 pulses vs Control	0.94 (0.75 to 1.17)	-6%	0.850
Stevens	2024	3 pulses vs Control	1.02 (0.81 to 1.28)	+2%	0.995
Stevens	2024	4 pulses vs Control	0.90 (0.71 to 1.12)	-10%	0.555
Stevens	2025	1 pulse vs Control	0.97 (0.77 to 1.21)	-3%	0.966
Stevens	2025	2 pulses vs Control	1.10 (0.87 to 1.39)	+10%	0.714
Stevens	2025	3 pulses vs Control	0.88 (0.69 to 1.11)	-12%	0.454
Stevens	2025	4 pulses vs Control	0.66 (0.52 to 0.84)	-34%	< 0.001

The loss is concentrated in 2025 and in Mullica Queen. Four pulses cut the expected count of Mullica Queen

in 2025 to 0.60 of Control, a fall of 40%, and three pulses to 0.70. In 2024 the four heat groups are closer together, and only two pulses in Mullica Queen reach the 5% level after adjustment.

5.3 Answer

Heat stress reduces the number of healthy berries, but the size of the reduction varies with cultivar and year. On average, heated plots produced 16% fewer healthy berries than Control, with a 95% confidence interval from 10% to 21% ($p < 0.001$). The 2025 season carries most of the loss, reaching 40% fewer healthy berries with four pulses in Mullica Queen and 34% in Stevens, whereas in 2024 the four heat groups sit closer together.

The joint tests show a clear heat effect, $p < 0.001$, and a large cultivar-by-year shift, $p < 0.001$. The heat effect also differs between years, $p = 0.009$, which is the source of the 2025 losses. It differs less clearly between cultivars, $p = 0.096$. The full three-way interaction does not improve the fit, $p = 0.263$; the comparison of the candidate models is in Appendix C, together with a beta-binomial model that treats the same data directly as a proportion and gives the same ordering.

6) How does heat stress affect yield as measured by the percentage of healthy fruit by weight?

The second percentage measure uses weight instead of count. For each plot, `healthy_weight_prop` is the weight of healthy fruit divided by total fruit weight, so it lies between 0 and 1. A beta GLMM with a logit link is the natural model for a proportion, with a random intercept for each pair and fixed effects for heat group, cultivar and year.

6.1 Final model

The model is

```
healthy_weight_prop ~ heat_group + cultivar + year + (1 | pair_id)
```

No plot has a share of exactly 0 or 1, so the raw proportions are used without adjustment. On the logit scale a coefficient is a change in log odds, and `exp(beta)` is the multiplicative change in the odds that a gram of harvested fruit is healthy, again against Control in 2024 for Mullica Queen.

6.2 Results

Table 5: Estimated share of healthy fruit by weight in each heat group, averaged over cultivars and years, from the final beta model, with 95% confidence intervals.

Heat group	Estimated share of healthy fruit by weight
Control	97.4% (96.4% to 98.2%)
1 pulse	87.6% (83.9% to 90.5%)
2 pulses	89.8% (86.3% to 92.5%)
3 pulses	90.7% (87.5% to 93.1%)
4 pulses	91.9% (89.0% to 94.1%)

Table 6: Effect of heat on the share of healthy fruit by weight, expressed as odds ratios relative to Control, adjusted for four comparisons with the same Control using Dunnett's method.

Comparison	Odds ratio (95% CI)	p
1 pulse vs Control	0.19 (0.10 to 0.34)	< 0.001
2 pulses vs Control	0.23 (0.14 to 0.38)	< 0.001
3 pulses vs Control	0.26 (0.15 to 0.44)	< 0.001
4 pulses vs Control	0.30 (0.16 to 0.56)	< 0.001

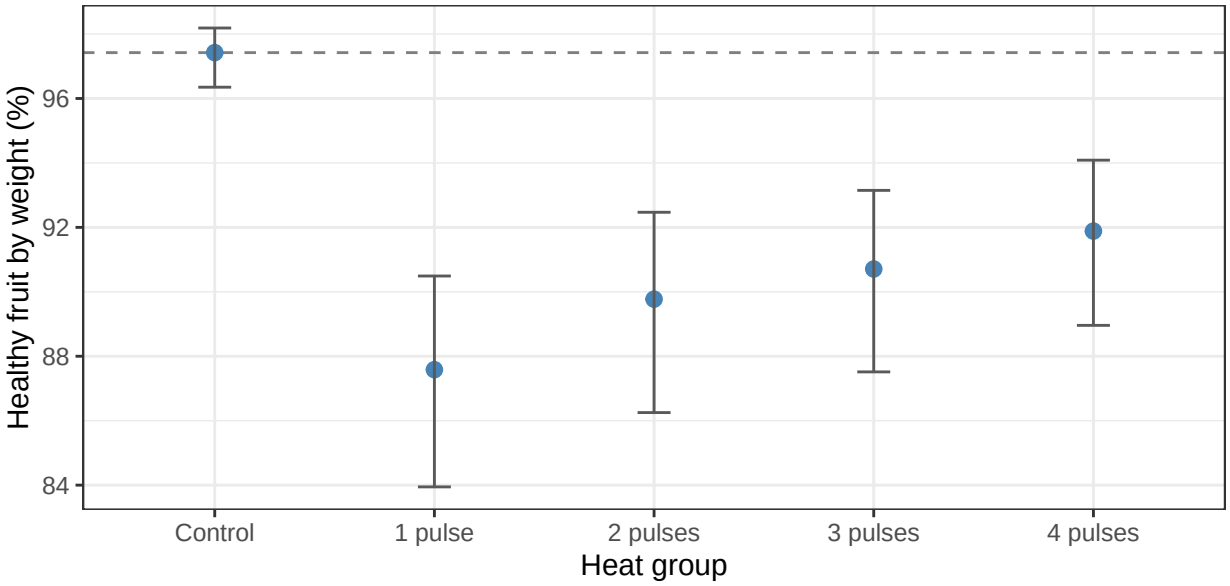


Figure 6: Estimated share of healthy fruit by weight for each heat group, averaged over cultivars and years, with 95% confidence intervals. The dashed line marks the Control estimate.

Control plots keep 97.4% of their harvest weight as healthy fruit. Every heated group is below that, and one pulse is the lowest at 87.6%. The four heat groups average 90.0%, about 7.4 percentage points below Control. The odds that a gram is healthy are 0.30 times the Control odds with four pulses and 0.19 times with one.

6.3 Answer

The share of healthy fruit by weight is highest in Control plots, 97.4%, and falls to 87.6% with one pulse. All four heat groups sit between 87.6% and 97.4%, well below Control, and each difference is clear at $p < 0.001$.

The number of pulses does not order the loss. One and two pulses sit below three and four pulses, so the damage appears once the plots are heated at all rather than growing with each pulse. Year makes no difference for this measure, $p = 0.937$, while the cultivars differ.

7) Conclusion

Heat stress lowered both the amount and the quality of the cranberry harvest. Averaged over the four heat intensities, total fruit weight was about 12% below Control (Section 3), healthy fruit weight was 20% below

(95% CI 14% to 25%, Section 4), and the expected number of healthy berries was 16% below (Section 5). Because the healthy measures fall faster than total weight, heat does not only make the crop smaller; it shifts the composition of the harvest towards rotten fruit.

The damage is concentrated in 2025. Four pulses cut total weight by about 37% and healthy weight by about 40% in that year, and cut the expected count of healthy berries by 40% in Mullica Queen. In 2024 the heat groups sit closer to Control and the ordering with the number of pulses is weaker. For the share of healthy fruit by weight the picture differs: Control plots keep 97.4% of their harvest weight as healthy fruit, one pulse is already the lowest at 87.6%, and extra pulses do not add to the loss (Section 6). With two years of data we cannot say whether the 2025 pattern or the 2024 pattern is more typical.

Mullica Queen and Stevens differ in level more than in sensitivity. Their baselines differ by a wide margin, and the 2025 Stevens baseline is much lower than in 2024, but there is no clear evidence that the proportional effect of heat differs between the cultivars. The same proportional loss therefore costs Mullica Queen more grams.

Three limitations apply. Only the heat treatment was assigned by the experimenters, so only its effects can be read as causal. Each cultivar grew in a single field, so cultivar differences cannot be separated from field differences. With three pairs per heat group in each cultivar and year, comparisons of single heat groups are imprecise, and the models were chosen after examining the data.

8) Reference

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9) Appendix

Appendix A: Total weight of fruit (Section 3)

A.1 Model specification The final model for plot i is

$$\log(Y_i) = \mu + \alpha_{g(i)} + \beta_{t(i)} + (\alpha\beta)_{g(i)\,t(i)} + \gamma_{c(i)} + (\beta\gamma)_{t(i)\,c(i)} + b_{k(i)} + \varepsilon_i,$$

where Y_i is the total fruit weight in grams and log is the natural logarithm. μ is the overall intercept, α_g the effect of heat group g, β_t the effect of year t, γ_c the effect of cultivar c, $(\alpha\beta)_{gt}$ the heat-by-year interaction, and $(\beta\gamma)_{tc}$ the cultivar-by-year shift. b_k is the random intercept of pair k, with $b_k \sim N(0, \sigma_b^2)$, and ε_i is the residual error, with variance $\sigma^2_{\text{Control}}$ in Control plots and σ^2_{Heated} in heated plots. The estimated standard deviations are $\hat{\sigma}_b = 0.067$, $\hat{\sigma}_{\text{Control}} = 0.09$ and $\hat{\sigma}_{\text{Heated}} = 0.164$. The model was fitted by restricted maximum likelihood in R with nlme. The estimated coefficients are in Section 3.2.

A.2 Choice of response scale and model **Response scale.** Total weight is strictly positive and right-skewed, so the response is logged.

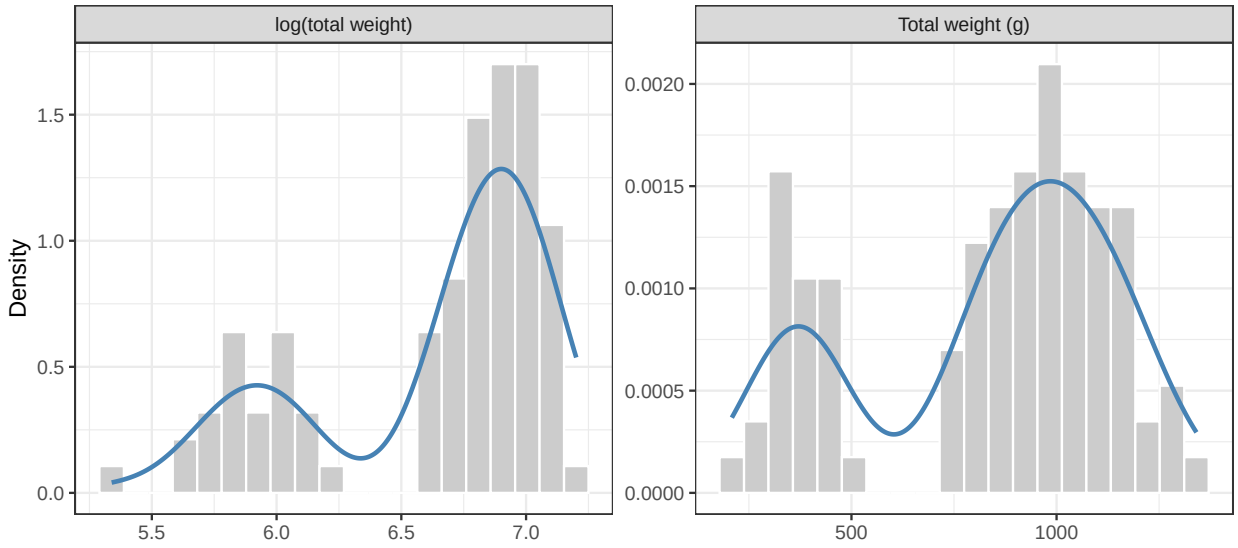


Figure 7: Total fruit weight on the raw scale (left) and the log scale (right). The raw weights are right-skewed; the log scale is closer to symmetric.

Across the 20 cells defined by cultivar, year and heat group, the Spearman correlation between the group mean and the group standard deviation is 0.49 on the raw scale and -0.15 on the log scale, so the log scale gives the more even spread.

Logging also makes each heat effect a ratio, so a ratio of 0.80 is 20% less fruit.

Choosing the model. Five models differ only in how heat enters the fixed part, and all five keep the random intercept for pair and the cultivar-by-year shift. H0 has one heat effect. H1 lets it differ by cultivar, H2 by year, and H3 includes both interactions. H4 replaces the five-level heat group by treatment by intensity, which splits the Control plots by the intensity of their paired heated plot instead of pooling them, so it tests whether pooling is justified.

The profiles below motivate the heat-by-year term: in 2024 the heat groups are close to Control, whereas in 2025 the mean falls with the number of pulses, and Stevens is much lower in 2025 than in 2024.

The candidate fixed-effect structures are:

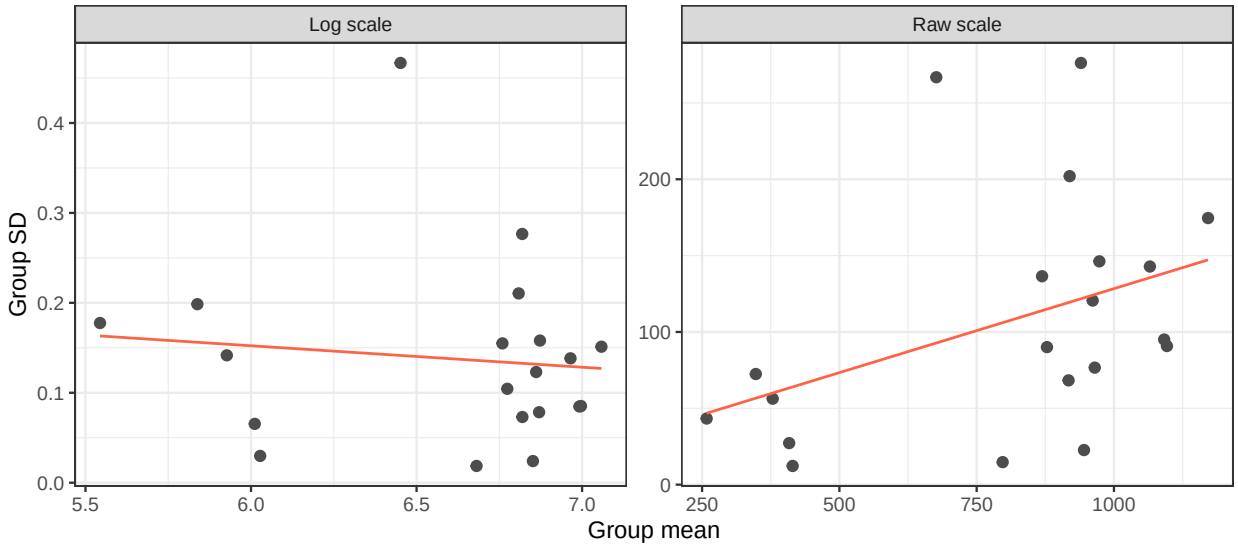


Figure 8: Group standard deviation against group mean on the raw scale (left) and the log scale (right). Each point is one cultivar, year and heat group; the line is a least-squares fit.

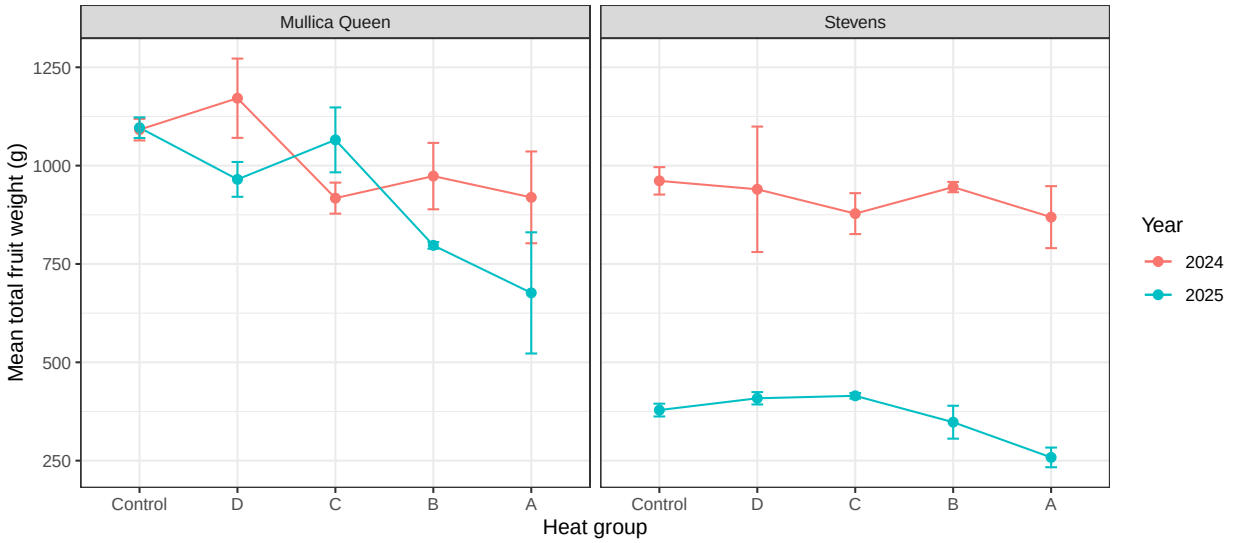


Figure 9: Mean total fruit weight by heat group and year, separately for each cultivar. In 2024 the four heat groups are close to Control; in 2025 the mean falls with the number of pulses, and Stevens is much lower than in 2024.

Table 7: AIC comparison for total weight (ML fits).

Model	df	AIC	dAIC
H0	10	-71.59	13.01
H1	14	-68.64	15.96
H2	14	-84.60	0.00
H3	18	-83.69	0.91
H4	13	-66.65	17.95

```
## $cult
## Data: data
## Models:
## tw_h0: log(total_weight) ~ heat_group + cultivar + year + year:cultivar + (1 | pair_id)
## tw_h1: log(total_weight) ~ heat_group * cultivar + year + year:cultivar + (1 | pair_id)
##       npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## tw_h0   10 -71.594 -45.951 45.797   -91.594
## tw_h1   14 -68.637 -32.736 48.318   -96.637 5.0426  4      0.283
##
## $year
## Data: data
## Models:
## tw_h0: log(total_weight) ~ heat_group + cultivar + year + year:cultivar + (1 | pair_id)
## tw_h2: log(total_weight) ~ heat_group * year + cultivar + year:cultivar + (1 | pair_id)
##       npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## tw_h0   10 -71.594 -45.951 45.797   -91.594
## tw_h2   14 -84.599 -48.698 56.300  -112.599 21.005  4  0.0003159 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $both
## Data: data
## Models:
## tw_h0: log(total_weight) ~ heat_group + cultivar + year + year:cultivar + (1 | pair_id)
## tw_h3: log(total_weight) ~ heat_group * (cultivar + year) + year:cultivar + (1 | pair_id)
##       npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## tw_h0   10 -71.594 -45.951 45.797   -91.594
## tw_h3   18 -83.688 -37.530 59.844  -119.688 28.094  8  0.0004567 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $year_after_cult
## Data: data
## Models:
## tw_h1: log(total_weight) ~ heat_group * cultivar + year + year:cultivar + (1 | pair_id)
## tw_h3: log(total_weight) ~ heat_group * (cultivar + year) + year:cultivar + (1 | pair_id)
##       npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## tw_h1   14 -68.637 -32.736 48.318   -96.637
## tw_h3   18 -83.688 -37.530 59.844  -119.688 23.052  4  0.0001237 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $cult_after_year
## Data: data
```

```
## Models:
## tw_h2: log(total_weight) ~ heat_group * year + cultivar + year:cultivar + (1 | pair_id)
## tw_h3: log(total_weight) ~ heat_group * (cultivar + year) + year:cultivar + (1 | pair_id)
##       npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## tw_h2   14 -84.599 -48.698 56.300   -112.60
## tw_h3   18 -83.688 -37.530 59.844   -119.69 7.0888  4     0.1313
```

H2 has the smallest AIC, so we take H2. The likelihood-ratio tests agree: heat by year improves the fit at p = <0.001, heat by cultivar does not at p = 0.283, and adding heat by cultivar after heat by year leaves the fit unchanged at p = 0.131. H4 has a larger AIC than H2, so the Control plots can be pooled into one Control group.

A.3 Residual variance and model checks The selected model assumes one residual variance. Control and heated plots may differ in spread, so the same fixed and random effects were refitted in nlme with a separate residual variance for each group:

Residual spread. With one residual variance the residuals of heated plots are more spread out than those of Control plots. Ignoring this would make comparisons of heated plots look more precise than they are. The likelihood-ratio test of the extra variance gives p = 0.00464, and the estimated residual standard deviation of heated plots is 1.8 times that of Control plots.

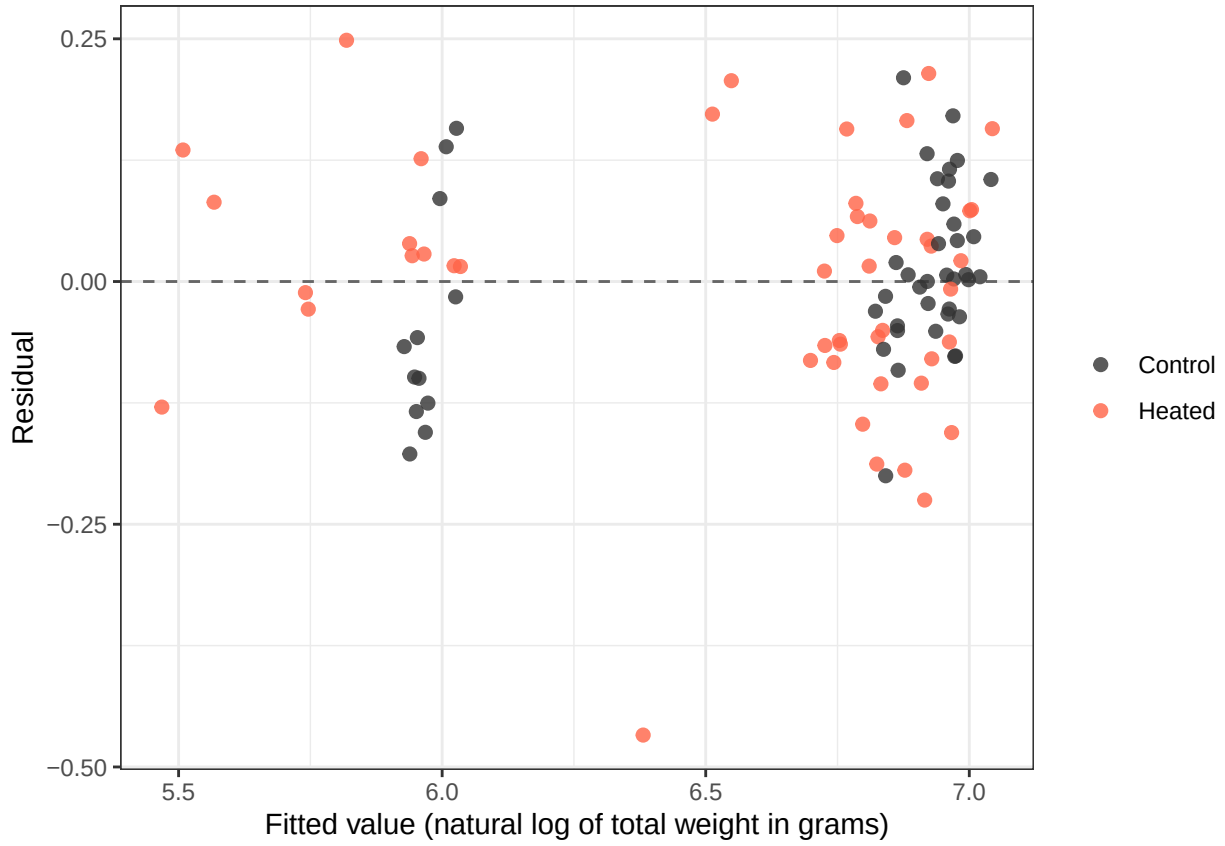


Figure 10: Residuals against fitted values from the selected model fitted with a single residual variance for all plots. Each point is one plot; black points are Control plots and red points are heated plots. The dashed line marks zero. The residuals of heated plots are clearly more spread out than those of Control plots.

After the refit the Pearson residuals and the estimated pair effects are close to normal.

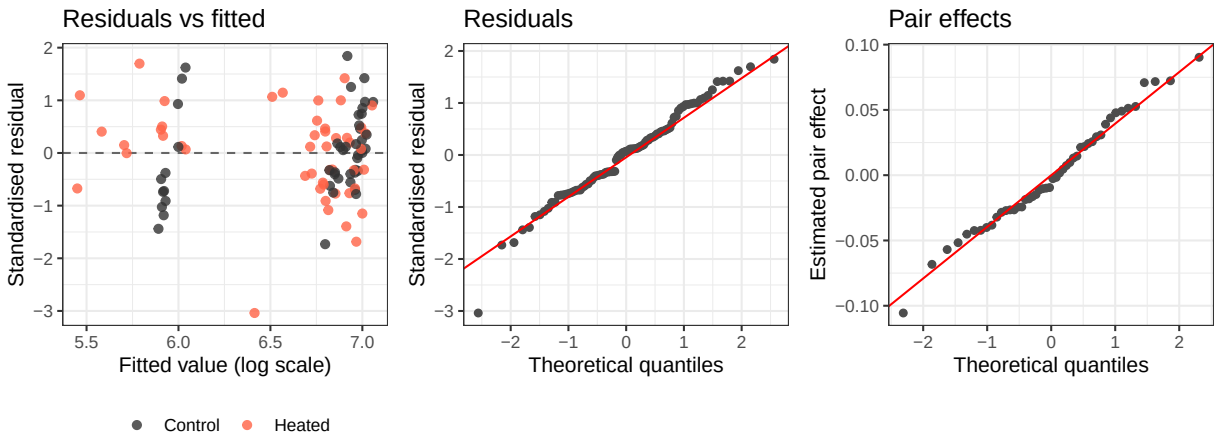


Figure 11: Diagnostic plots for the final model, which allows separate residual variances for heated and Control plots. Left: standardised (Pearson) residuals against fitted values; black points are Control plots and red points are heated plots, and the dashed line marks zero. Middle: normal quantile plot of the standardised residuals. Right: normal quantile plot of the estimated pair effects. Points close to the red lines indicate approximately normal distributions.

A.4 Sensitivity analysis We removed the 3 pairs that hold the three largest standardised residuals, 6 plots in all, and refitted the final model. The average reduction over the four heat groups is 7% in 2024 and 13% in 2025, against 8% and 15% for the full data, so the conclusions do not depend on these plots.

Appendix B: Total weight of healthy fruit (Section 4)

B.1 Model specification

The final model for plot i is

$$\log(Y_i) = \mu + \alpha_{g(i)} + \beta_{t(i)} + (\alpha\beta)_{g(i)\,t(i)} + \gamma_{c(i)} + (\beta\gamma)_{t(i)\,c(i)} + b_{k(i)} + \varepsilon_i,$$

where Y_i is the healthy fruit weight in grams, log is the natural logarithm, μ is the overall intercept, α_g is the effect of heat group g (Control or 1, 2, 3 or 4 pulses), β_t is the effect of year t, γ_c is the effect of cultivar c, $(\alpha\beta)_{gt}$ allows the heat effect to differ between years, and $(\beta\gamma)_{tc}$ allows the cultivar baseline to differ between years. The term b_k is the random effect of pair k, with $b_k \sim N(0, \sigma_b^2)$, and ε_i is the residual error, with $\varepsilon_i \sim N(0, \sigma^2_{\text{Control}})$ for Control plots and $\varepsilon_i \sim N(0, \sigma^2_{\text{Heated}})$ for heated plots. The estimated standard deviations are $\hat{\sigma}_b = 0.062$, $\hat{\sigma}_{\text{Control}} = 0.093$ and $\hat{\sigma}_{\text{Heated}} = 0.212$. The model was fitted by restricted maximum likelihood with the nlme package in R. The estimated coefficients are listed in Table 9 (Appendix B.4).

B.2 Choice of response scale and model

Response scale. We modelled the natural logarithm of healthy weight. On the original scale, groups with a higher mean healthy weight also varied more (Spearman correlation between group mean and group standard deviation 0.44 across the 20 combinations of cultivar, year and heat group), which violates the model's assumption of equal variance; on the log scale this association was much weaker (-0.22).

Choosing the model. We compared six candidate models that differed only in how the heat effect was allowed to vary between cultivars and years (Table 8), using likelihood-ratio tests and the Akaike and

Bayesian information criteria (AIC and BIC). There was strong evidence that heat affected healthy weight ($p < 0.001$). Allowing the heat effect to differ between years improved the fit ($p = 0.0064$), whereas allowing it to differ between cultivars did not ($p = 0.29$). We therefore selected the model with a year-specific heat effect, which also had the lowest AIC; BIC, which penalises additional parameters more heavily, preferred the simpler model with a common heat effect.

Table 8: Candidate models for the natural logarithm of healthy fruit weight. All models include cultivar, year, the cultivar-by-year interaction and a random effect for each pair, and were fitted by maximum likelihood with a single residual variance. Parameters counts all estimated quantities, including the variances. Lower AIC and BIC indicate a better balance between fit and model size. The p-value is from a likelihood-ratio test against the model in the Compared with column.

Model	Heat effect	Parameters	AIC	BIC	Compared with	p-value
1	None	6	-11.8	3.6		
2	Same in all cultivars and years	10	-46.9	-21.2	Model 1	< 0.001
3	Differs between cultivars	14	-43.8	-7.9	Model 2	0.290
4	Differs between years (selected)	14	-53.2	-17.3	Model 2	0.006
5	Differs between cultivars and between years	18	-51.4	-5.3	Model 4	0.179
6	Differs in every cultivar-year combination	22	-48.1	8.3	Model 5	0.324

B.3 Residual variance and model checks

Unequal variances in heated and Control plots. When the selected model was fitted with the same residual variance for all plots, the residuals of heated plots were clearly more spread out than those of Control plots (Figure 12). Ignoring this would understate the uncertainty of comparisons involving heated plots, so we refitted the model with separate residual variances for heated and Control plots, which fitted much better (likelihood-ratio test, $p < 0.001$). The estimated residual standard deviation of heated plots was about 2.3 times that of Control plots. With this change, the residuals had a similar spread in both groups and both the residuals and the estimated pair effects were close to normally distributed (Figure 13), apart from two heated plots with unusually low healthy weight, which we examine in Appendix B.5.

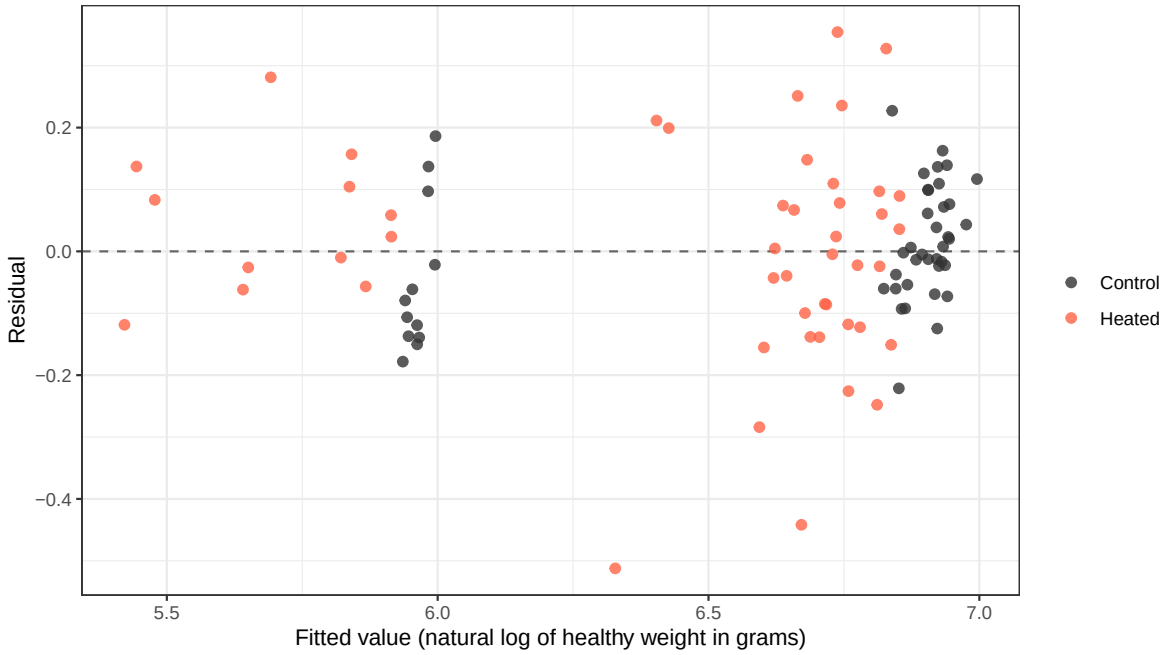


Figure 12: Residuals against fitted values from the selected model fitted with a single residual variance for all plots. Each point is one plot. Black points are Control plots and red points are heated plots. The dashed line marks zero. The residuals of heated plots are clearly more spread out than those of Control plots.

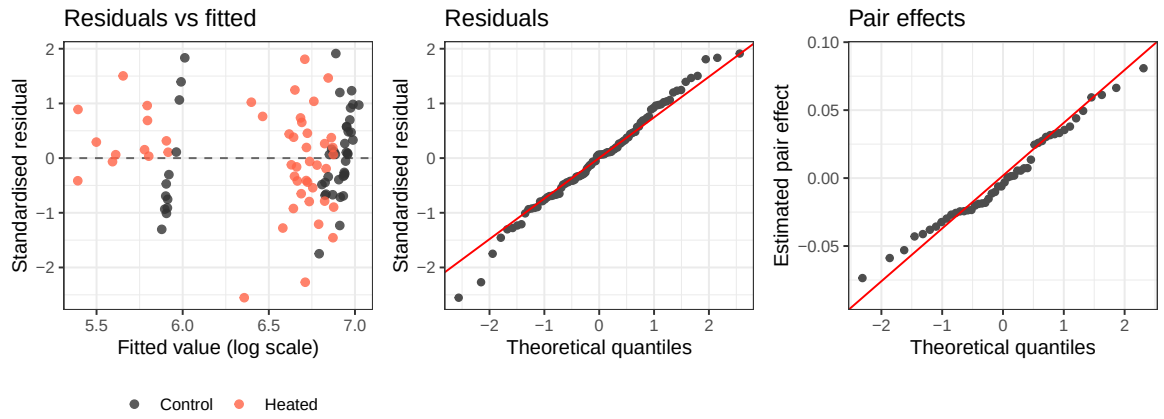


Figure 13: Diagnostic plots for the final model, which allows separate residual variances for heated and Control plots. Left: standardised (Pearson) residuals against fitted values. Black points are Control plots and red points are heated plots, and the dashed line marks zero. Middle: normal quantile plot of the standardised residuals. Right: normal quantile plot of the estimated pair effects. Points close to the red lines indicate approximately normal distributions.

B.4 Coefficients

Table 9: Estimated fixed effects of the final model. Estimates are on the natural log scale. Ratio is the exponential of the estimate, i.e. the multiplicative change in healthy weight. The reference levels are Control, 2024 and Mullica Queen. For the intercept, the exponential is the estimated healthy weight in grams of Control plots of Mullica Queen in 2024. The heat group terms compare each heat group with Control in 2024, the terms marked × 2025 give the additional multiplicative change in 2025.

Term	Estimate (log scale)	Standard error	Ratio	95% CI for ratio	p-value
Intercept (Control, 2024, Mullica Queen)	6.956	0.031	1049.56	985.47 to 1117.83	< 0.001
1 pulse	-0.178	0.091	0.84	0.70 to 1.01	0.057
2 pulses	-0.221	0.091	0.80	0.67 to 0.96	0.019
3 pulses	-0.108	0.091	0.90	0.75 to 1.08	0.239
4 pulses	-0.227	0.091	0.80	0.66 to 0.96	0.016
Year 2025	-0.012	0.044	0.99	0.90 to 1.08	0.791
Stevens	-0.105	0.043	0.90	0.83 to 0.98	0.018
1 pulse × 2025	0.061	0.128	1.06	0.82 to 1.38	0.638
2 pulses × 2025	0.146	0.128	1.16	0.89 to 1.50	0.262
3 pulses × 2025	-0.201	0.128	0.82	0.63 to 1.06	0.125
4 pulses × 2025	-0.287	0.128	0.75	0.58 to 0.97	0.031
2025 × Stevens	-0.908	0.061	0.40	0.36 to 0.46	< 0.001

B.5 Sensitivity analysis

To check whether our conclusions depend on the two heated plots with unusually low healthy weight, we removed the 2 pairs containing them (4 plots, 4.2% of the data) and refitted the final model. The estimated effects became slightly smaller, but all main conclusions were unchanged: the average reduction was 14% in 2024 and 20% in 2025, and the reduction per additional pulse in 2025 was 10%. The only change was that two heat groups in 2024 became individually statistically discernible, which confirms that the evidence for individual heat groups in 2024 is borderline.

Appendix C: Healthy count: model building (Section 5)

Counts are non-negative with a variance that grows with the mean, so a negative-binomial model is used rather than a Poisson or a normal model. The three candidate fixed-effect structures share the random intercept for pair and were fitted with maximum likelihood:

Table 10: Candidate models for the number of healthy berries. All models include a random intercept for each pair. Lower AIC indicates a better balance between fit and model size.

Model	Parameters	AIC
Main effects	9	1175.9
All two-way interactions	18	1106.5
All three-way interactions	22	1109.3

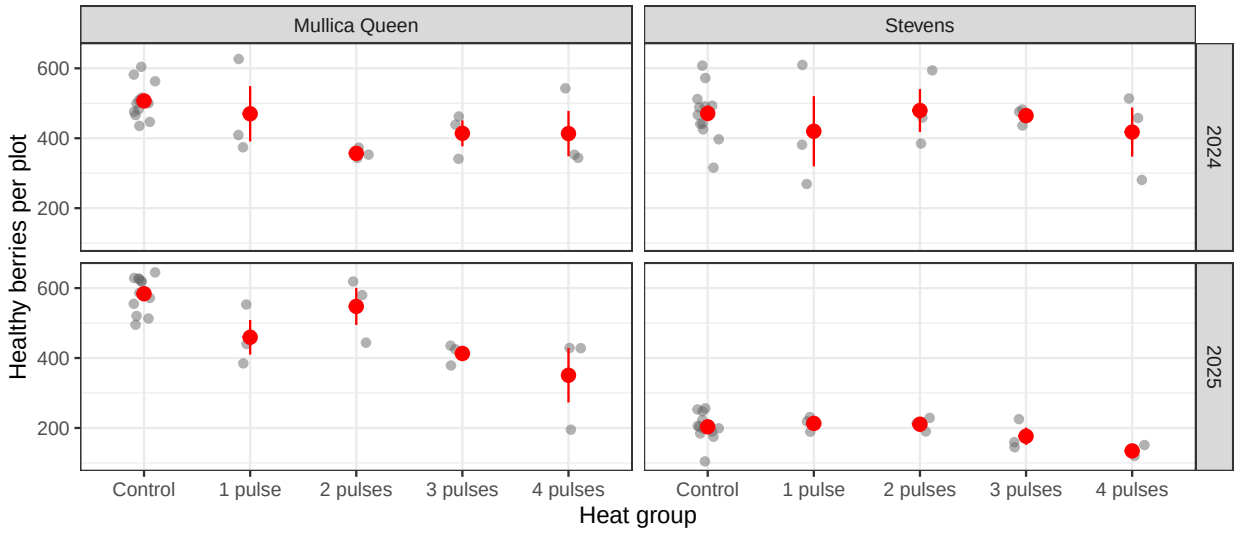


Figure 14: Number of healthy berries per plot by heat group, with the mean and one standard error in red. Grey points are individual plots.

Table 11: Likelihood-ratio tests between the candidate models for the number of healthy berries.

Comparison	Chi-square	df	p
Main effects to all two-way interactions	87.4	9	< 0.001
All two-way to all three-way interactions	5.2	4	0.263

Adding the two-way interactions improves the fit, $p < 0.001$, while adding the three-way interaction does not, $p = 0.263$, and it raises the AIC. The model with all two-way interactions is therefore used for the estimates in Section 5.

A proportion model as a check. The section models counts, but the measure is a percentage, so the same data were also modelled directly as a proportion. A beta-binomial GLMM was fitted to the number of healthy and rotten berries of each plot, with a logit link, the same fixed effects and the same random intercept for pair:

Table 12: Estimated share of healthy berries by count in each heat group, averaged over cultivars and years, from a beta-binomial GLMM on the healthy and rotten counts, with 95% confidence intervals.

Heat group	Estimated share of healthy berries by count
Control	95.1% (93.8% to 96.1%)
1 pulse	84.1% (80.0% to 87.5%)
2 pulses	84.5% (80.4% to 87.9%)
3 pulses	84.1% (80.0% to 87.6%)
4 pulses	82.1% (77.7% to 85.8%)

The two models answer different questions. The negative-binomial model describes how many healthy berries a plot carries, while the beta-binomial model describes the share of the berries that are healthy once the total is known. Their likelihoods therefore refer to different data, so AIC cannot choose between them. On

the pattern the two agree: Control keeps 95.1% of its berries healthy, every heat group is lower, and four pulses are the lowest at 82.1%, which is the same ordering as the count model in Section 5.

Appendix D: Share of healthy fruit by weight: model building (Section 6)

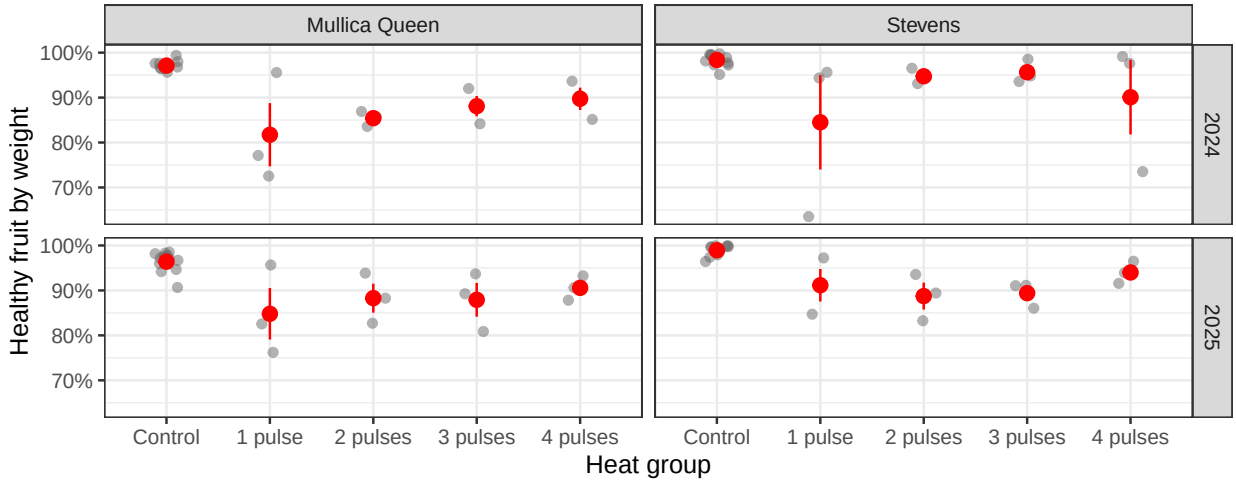


Figure 15: Share of healthy fruit by weight for each plot, by heat group, with the mean and one standard error in red. The share is bounded by 0 and 1, which is why a beta model is used.

The share is a proportion bounded by 0 and 1, and its variance depends on its mean. None of the 96 plots sits exactly at 0 or 1; the observed shares run from about 64% to nearly 100%. A beta GLMM on the logit scale is used, with a random intercept for pair.

The beta-binomial model on the counts of healthy and rotten berries is an alternative way to describe the same composition, but it models a share of berries by number, not a share of weight, so its likelihood is not comparable with the beta model. The beta model on the proportion is used for the results in Section 6.

Year contributes little to the share by weight. Dropping it changes the fit by $p = 0.937$, so the year term is kept only so that this model has the same fixed-effect structure as the other yield models and the measures stay comparable.